

Pembroke-Graham's Pond

Shoreline length: 12 km



Shore Types

Pembroke-Graham's Pond is an exposed coast along the east end of PEI, with a majority of wetlands, low plains, and dunes.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
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18	2	2	12	41	5	17	3
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Governance, Infrastructure, and Economy

Pembroke-Graham's Pond SMU includes a portion within the Town of Three Rivers (~19%), with the remaining shoreline being entirely unincorporated and subject to provincial planning authority.

Coastal infrastructure includes the Graham's Pond Harbour, along with Graham's Pond Road (Gaspereaux), Route 17 at Graham's Pond, and MacLure Pond Road (Gaspereaux).

Coastal economic drivers include both Gaspereaux Harbour and the Graham's Pond wharf support active fisheries. Local and legacy land use constraints include a few small coastal lots and unpaved coastal roads.



Graham's Pond Harbour

Natural Assets and Heritage

Popular Poverty Beach is located at the south end of the SMU

The shoreline includes critical habitat for bird species including the endangered Piping Plover.



Irvings Cape (left), Poverty Beach and down-drift sandspit (right) at south end of SMU

Planning and Policy Framework

Where provincial rules apply, land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted. Within Three Rivers, the Town's municipal rules apply.

Development on the coast is informed by the Environmental Protection Act, including a 15m buffer, with no additional building

setbacks aside from the buffer. The Official Plan includes specific guidance permitting armouring, though such provisions are secondary to any provincial standards, and includes some policies regarding expansion of buildings within the sea level rise overlay. There are no specific flood elevations or vertical setbacks established, but the Official Plan does include specific erosion-based setbacks, with the possibility of reduced setbacks under certain conditions. Hazards are assessed at the subdivision stage.

Coastal Flood and Erosion Risk

While low plains, dunes and wetland shores are vulnerable to increasing flood hazards, infrastructure risk is low. Erosion risk is low due to the sparse amount of infrastructure and buildings.

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential shoreline change or salt marsh migration to future time horizons.

Number of at-risk buildings [N. per km shoreline]

	Present	0.3m SLR (2050)	0.6m SLR (2080)	1.2m SLR (2130)
SMU Pembroke-Graham's Pond				
Flood risk	45 [2/km]	47 [2/km]	48 [2/km]	54 [3/km]
Erosion risk	N/A	5 [<1/km]	7 [<1/km]	26 [1/km]
PU Grahams Pond Harbour				
Flood risk	44 [33/km]	46 [35/km]	46 [35/km]	46 [35/km]
Erosion risk	N/A	2 [2/km]	4 [3/km]	21 [16/km]

Sediment transport - The net sediment transport across the SMU is high and flows southward towards Poverty Beach and its terminal sandspit, which provides partial shelter to Murray Harbour North.

Proposed SMP Strategies

The primary strategy for Pembroke-Graham's Pond is No Active Intervention, which will preserve natural shorelines and sediment supply to beaches. For infrastructure at risk, managed realignment is proposed, as well as resist for Grahams Pond Harbour.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
Proposed SMU strategy	NAI	MR	MR
Rationale	Mid- to long-term MR will reduce risk to the limited number of vulnerable assets while preserving natural shorelines, including dunes, wetlands and cliff habitat.		
Proposed PU strategy			
Grahams Pond Harbour	R	R	R
Rationale	Resist strategy is warranted for the harbour as critical infrastructure to the community.		



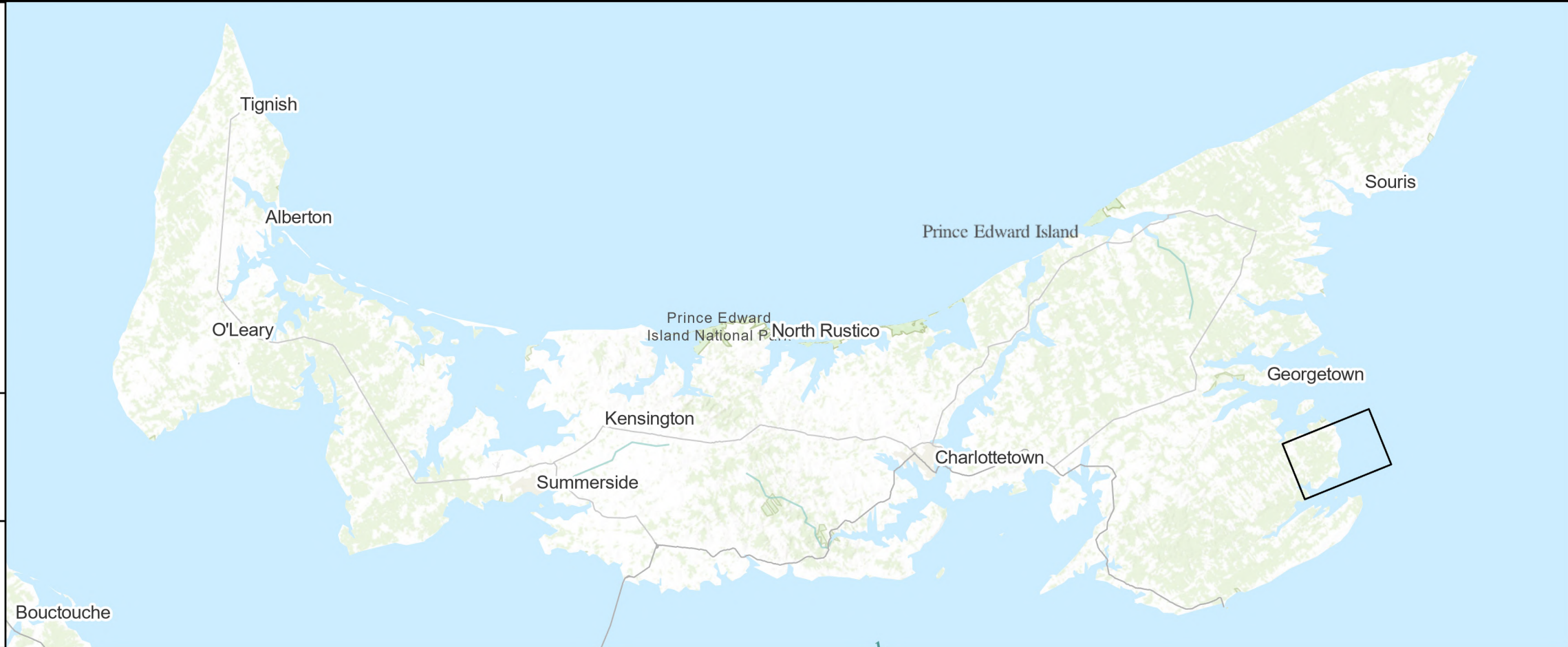
LEGEND

- Policy Unit
- Shoreline Management Unit
- Net Longshore Sediment Transport
- High Hazard Area
- Flood Extent - Present Day (100-year Return)
- Flood Extent - 1.2m SLR (100-year Return)
- Property Boundary
- Building > 90ft²

0 200 400 800
Metres

Coordinate System:
NAD 83 CSRS PEI Double Stereographic

NOTES:



**COASTAL EROSION
AND FLOOD HAZARD**

**Shoreline Management Unit:
Pembroke-Grahams Pond**

Prince Edward Island
Île-du-Prince-Édouard